High-Functioning Older Adults Use Greater Muscle Contributions to Whole-Body Angular Momentum than Younger Adults to Regulate Balance

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Introduction: Balance is maintained by regulating the whole-body angular momentum (*H*) about the center-of-mass [1]. Older adults (OLDER) often have larger peak-to-peak ranges of *H* than younger adults (YOUNG), which indicates poorer balance control [2] and increased fall risk. Muscles regulate *H* by contributing to the net external moment (NetExtM) about the center-of-mass [3]. Identifying age-related changes in muscle contributions to the NetExtM may provide insight into balance mechanisms in OLDER. The purpose of this study was to compare individual muscle contributions to *H* regulation between YOUNG and OLDER.

Methods: Motion capture data of 5 YOUNG (37.4±3.2 years) and 5 OLDER (71.8±1.6 years) walking at their preferred speed (YOUNG: 1.4±0.2 m/s; OLDER: 1.5±0.1 m/s) were used to generate simulations in OpenSim [4]. OpenSim MoCoInverse [5] estimated muscle forces while tracking electromyography from 10 muscles. Muscle forces were used to calculate muscle contributions to NetExtM [3] in frontal and sagittal planes. The *H* range and integral of each muscle's contribution to NetExtM over stance were normalized to height, weight, and speed and compared between age-groups using independent samples t-tests ($\alpha<0.05$).

Results: *H* range did not differ in the sagittal plane; yet, OLDER had larger contributions to forward *H* from the medial hamstrings and gastrocnemius, and larger contributions to backward *H* from the rectus femoris, vasti, and tibialis anterior than YOUNG (Fig. A-B; $p<0.05$). OLDER had lower frontal *H* range than YOUNG ($p=0.043$) and larger contributions to *H* towards the contralateral leg from the gluteus maximus, vasti, and gastrocnemius (Fig. C-D; $p<0.05$).

Discussion: Compared to YOUNG, OLDER used greater muscular effort to regulate similar sagittal plane and greater frontal plane H range. This greater muscular effort may be a cautious strategy to avoid falls by more tightly controlling balance regulation and may contribute to greater energy cost of walking in OLDER.

References: [1] Herr & Popovic, *J Exp Biol*, 2008. [2] Neptune & Vistamehr, *J Biomech Eng*, 2019. [3] Neptune & McGowan, *J Biomech*, 2011. [4] Delp, *IEEE Trans Biomed Eng*, 2007. [5] Dembia, *PLoS Comp Bio*, 2020.

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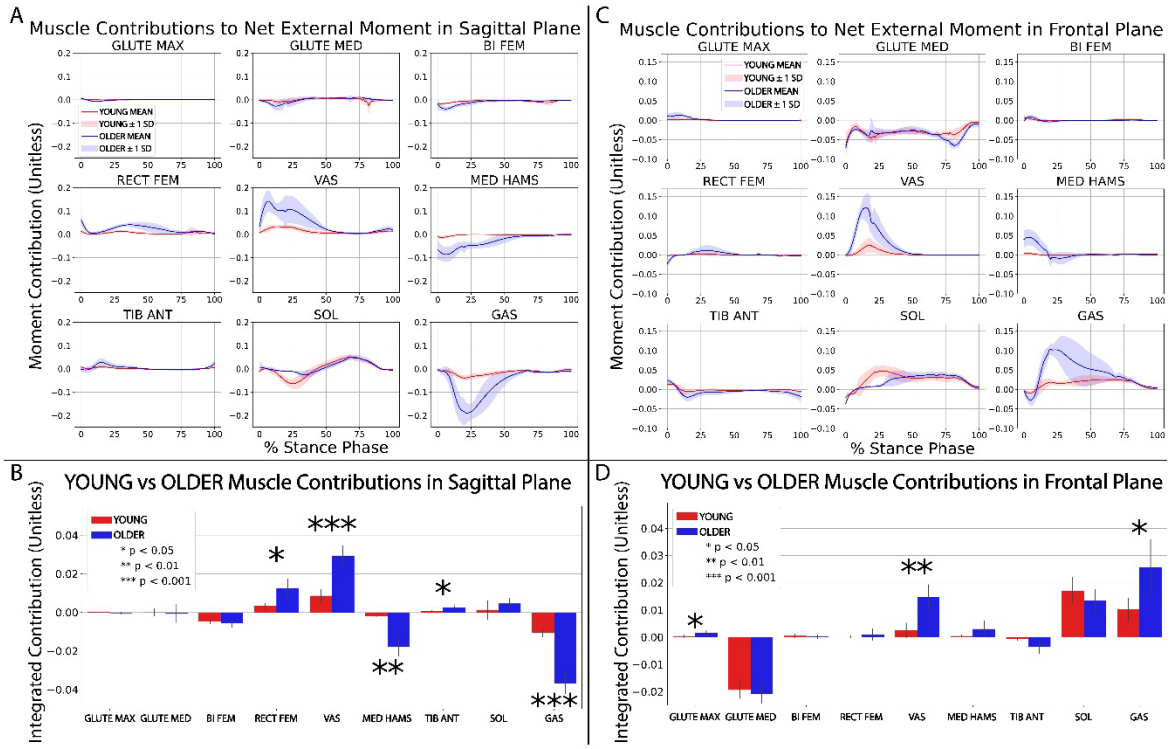

Fig. 1: Muscle contributions to balance regulation in younger and older adults. A) Muscle contributions to the sagittal NetExtM. B) Integrated muscle contributions to the sagittal NetExtM. C) Muscle contributions to the frontal NetExtM. D) Integrated muscle contributions to the frontal NetExtM. In the sagittal plane, positive (negative) values indicate the muscle is producing backward (forward) H. In the frontal plane, positive (negative) values indicate the muscle is producing H toward the contralateral (ipsilateral) limb. Muscle contributions from gluteus maximus and gluteus medius lines of action were summed in GLUTE MAX and GLUTE MED, respectively. Muscle contributions from semimembranosus and semitendinosus were summed in MED HAMS, from medial and lateral gastrocnemius were summed in GAS, and from vastus medialis, lateralis, and intermedius were summed in VAS. BI FEM (biceps femoris long head), RECT FEM (rectus femoris), TIB ANT (tibialis anterior), and SOL (soleus) each represent contributions from one muscle.